World Model Implanting for Test-time Adaptation of Embodied Agents

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Abstract

Embodied agents driven by large language models (LLMs) often fail to generalize to novel tasks and environments, because the LLM's internal world knowledge cannot be efficiently updated for new domains without costly retraining. We propose WorMI (**Wor**ld **M**odel **I**mplanting), a plug-and-play framework for test-time adaptation of LLM-based embodied agents. WorMI maintains a library of independently trained, domain-specific world models and composes them with a frozen LLM policy at inference time. Relevant world models are selected via prototype-based retrieval using object-wise trajectory embeddings compressed by k-center clustering and matched with Wasserstein distance. The retrieved world models are fused through a hierarchical *world-wise compound attention* mechanism: a world-level cross-attention integrates multiple world representations, and a reasoning-level cross-attention aligns the fused knowledge with the LLM's internal representations. A lightweight composer is meta-learned with REPTILE across subsets of world models, enabling rapid specialization. World models can be implanted or removed at test time, supporting scalable, continual knowledge composition without modifying the base LLM. Extensive experiments on VirtualHome and ALFWorld show that WorMI achieves state-of-the-art zero-shot and few-shot performance, outperforming the strongest baseline by up to 20.41% in success rate on unseen tasks and scenes, while also excelling on challenging long-horizon and multi-instruction settings.

1. Introduction

The deployment of embodied agents in open-ended environments demands both broad commonsense reasoning and accurate, domain-specific knowledge of objects, affordances, and action dynamics. Large language models (LLMs) have demonstrated remarkable planning capabilities as zero-shot or few-shot planners for household tasks (Huang et al., 2022a; Song et al., 2023; Hazra et al., 2024; Singh et al., 2023; Liang et al., 2023), leveraging the vast world knowledge encoded in their parameters. However, the knowledge captured during pretraining is static and general: when an agent encounters a novel environment with unfamiliar objects or interaction rules, fine-tuning the entire LLM is prohibitively expensive, and prompting alone cannot inject structured, procedural domain knowledge reliably (Yao et al., 2022; Huang et al., 2022b).

A natural solution is to train *world models*—compact models that capture environment-specific transition dynamics and object interactions—and supply them to the agent at test time. Yet existing approaches either require end-to-end retraining whenever a new domain is encountered, or rely on simple prompting that cannot encode rich structural knowledge. The key challenge is therefore: *how can we efficiently inject domain-specific world knowledge into a frozen LLM policy on the fly?*

We address this challenge with WorMI (**Wor**ld **M**odel **I**mplanting), a framework that *implants* pre-trained domain world models into a frozen LLM reasoning policy at test time. WorMI treats each world model as a separately fine-tuned language model that encodes the dynamics, objects, and affordances of a particular domain. At inference time, given a target environment, WorMI (i) retrieves the most relevant world models from a library using prototype-based similarity matching, and (ii) fuses them with the LLM's hidden representations via a novel *world-wise compound attention* module.

The prototype-based retrieval builds object-wise state embeddings from domain trajectories, compresses them into compact k-center prototypes via k-means clustering, and matches the target environment using Wasserstein distance over prototype sets—yielding retrieval that is both efficient and robust to distributional variation within a domain.

The world-wise compound attention operates in two stages. First, a *world-level cross-attention* integrates the hidden representations of multiple retrieved world models, attending differentially to each world model to produce a unified world representation. Second, a *reasoning-level cross-attention* aligns this unified world representation with the frozen LLM's layer-wise hidden states, injecting world knowledge directly into the reasoning stream. A lightweight composer atop these attention modules is meta-learned with

REPTILE (Nichol, 2018) across different subsets of world models, enabling rapid specialization with minimal target-domain data.

Crucially, world models can be implanted or removed at any time without retraining the LLM, supporting continual and compositional knowledge management. This makes WorMI practically appealing: a new environment requires only training a small world model and registering it in the library, after which it can be immediately leveraged by the agent.

We evaluate WorMI on VirtualHome (Puig et al., 2018) and ALFWorld (Shridhar et al., 2021) in zero-shot and few-shot settings. Our method consistently and substantially outperforms strong baselines including SayCanPay (Hazra et al., 2024) and LLM-Planner (Song et al., 2023). Under the most challenging zero-shot, unseen-task/unseen-scene setting, WorMI achieves 66.12% success rate on VirtualHome ($+20.41\%$ over SayCanPay) and 51.67% on ALFWorld ($+12.01\%$). In few-shot settings, the gains are even larger ($+26.58\%$ on VirtualHome, $+19.16\%$ on ALFWorld on average). We further demonstrate the utility of WorMI on long-horizon and multi-instruction tasks, and analyze the effect of continual model implanting and removal.

Our contributions are:

1. We introduce WorMI, a test-time world model implanting framework that enables plug-and-play domain adaptation of LLM-based embodied agents without any LLM retraining.
2. We propose prototype-based retrieval using object-wise trajectory embeddings and k-center prototypes with Wasserstein distance for efficient, robust world model selection.
3. We design a world-wise compound attention module that hierarchically integrates multiple world models and aligns world knowledge with LLM reasoning; the composer is meta-learned for rapid adaptation.
4. Through extensive experiments, we demonstrate superior zero-shot and few-shot performance on VirtualHome and ALFWorld, alongside strong results in complex instruction and continual implanting scenarios.

2. Related Work

LLM-based planning for embodied agents. Large language models have been applied as high-level planners for embodied tasks in environments such as VirtualHome (Puig et al., 2018) and ALFWorld (Shridhar et al., 2021). Zero-shot planner (ZSP) (Huang et al., 2022a) showed that sufficiently large pretrained LLMs can decompose household tasks into actionable steps without task-specific fine-tuning. LLM-Planner (Song et al., 2023) demonstrated competitive few-shot planning using under 0.5% of paired training data. SayCanPay (Hazra et al., 2024) combines LLM-generated actions with learned affordance and cost functions via heuristic search for feasible and cost-efficient plans. ProgPrompt (Singh et al., 2023) structures LLM prompts as programs for situated planning, while Code-as-Policies (Liang et al., 2023) leverages code-writing LLMs for robot control. ReAct (Yao et al., 2022) and Inner Monologue (Huang et al., 2022b) incorporate environmental feedback into the reasoning process. DEPS (Wang et al., 2023) uses LLMs with interactive planning for long-horizon multi-task settings. While these methods excel in known domains, they do not provide a mechanism for efficiently injecting new domain knowledge into a frozen LLM at test time.

World models and model composition. World models learn to simulate environment dynamics and are central to model-based planning and reinforcement learning. In the context of language models, world knowledge is encoded implicitly in parameters and can be accessed through prompting, but cannot be easily updated or composed. CALM (Bansal et al., 2024) proposes to compose LLMs via cross-attention, augmenting a base LLM with domain-specific models while preserving the original capabilities. LLM-Blender (Jiang et al., 2023) ensembles multiple LLMs via pairwise ranking and generative fusion. FuseLLM (Wan et al.) studies knowledge fusion across models of different architectures. Mixture-of-Experts (Shazeer et al., 2016) learns to route tokens to specialized sub-networks. Unlike these approaches, WorMI is designed specifically for test-time adaptation of embodied agents, with retrieval-based selection and hierarchical attention that fuses world knowledge with LLM reasoning.

Meta-learning and fast adaptation. Meta-learning aims to obtain models that can quickly adapt to new tasks from few examples. MAML (Finn et al., 2017) learns a parameter initialization amenable to fast fine-tuning. REPTILE (Nichol, 2018) simplifies this to a first-order algorithm. Memory-augmented networks (Santoro et al., 2016) and learned optimizers (Andrychowicz et al., 2016) have also been explored. WorMI uses REPTILE to meta-learn the compound attention composer across diverse subsets of world models, enabling rapid specialization when a small amount of target-domain data is available.

Multi-agent and compositional LLM systems. Multi-agent frameworks such as AutoGen (Wu et al.) and MetaGPT (Hong et al.) decompose complex tasks across specialized LLM agents. Domain generalization methods (Zhou et al., 2022) address the challenge of out-of-distribution generalization, including via ensemble and

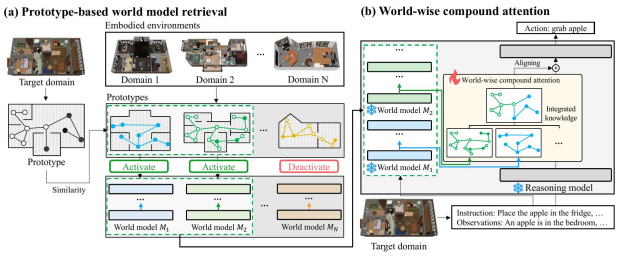


Figure 1: Overall procedure of WorMI. Domain-specific world models are trained independently, then retrieved and composed with a frozen LLM reasoning model at test time via world-wise compound attention.

meta-learning approaches. WorMI complements these directions by providing a principled mechanism for composing domain knowledge at the representation level.

Continual learning for embodied agents. Kim et al. (2024) propose continual learning setups for embodied agents learning new behaviors and environments. WorMI's implant/remove mechanism supports a similar incremental learning paradigm, enabling world models to be registered or deregistered as the agent encounters new or obsolete domains.

3. Approach

3.1. Problem Setup

We consider an embodied agent that executes natural language instructions in household environments. The agent follows a frozen LLM reasoning policy π_θ parameterized by weights θ. The policy maps a history of observations and instructions to a sequence of actions from a predefined action space. We assume access to a library of N pre-trained world models $\{\mathcal{W}_i\}_{i=1}^{N}$, each capturing the dynamics of a distinct domain (e.g., kitchen manipulation, bathroom tasks). At test time, we observe a target environment and must adapt the agent without retraining π_θ.

Overview. WorMI operates in three phases: (1) *world model training*, where individual world models are fine-tuned on domain trajectories; (2) *prototype-based retrieval*, where the most relevant world models are selected for the target environment; and (3) *world-wise compound attention*, where the retrieved world models are fused with the frozen LLM to produce an adapted policy. The overall procedure is illustrated in Figure 1.

3.2. World Model Training

Each world model $\mathcal{W}_i$ is a language model fine-tuned on a domain-specific dataset $\mathcal{D}_i$ consisting of task trajectories. The trajectories contain sequences of states, objects, and actions, which teach the world model the affordances, object relationships, and transition dynamics of the domain. We use behavior cloning (supervised fine-tuning) on next-token prediction as the training objective. World models can be of smaller capacity than the reasoning LLM, reducing the per-domain training cost. Each world model is trained independently and stored in the library alongside its domain dataset.

3.3. Prototype-Based World Model Retrieval

Given a target environment at test time, we wish to retrieve the k most relevant world models from the library. Naïvely computing similarity between raw trajectories is expensive and sensitive to superficial differences. Instead, WorMI builds compact *prototype representations* for each world model.

Prototype construction. For each world model $\mathcal{W}_i$, we process its domain dataset $\mathcal{D}_i$ through a sentence embedding model Φ to obtain object-wise state embeddings. Specifically, each trajectory observation is encoded by Φ, yielding a set of embeddings $\mathcal{E}_i = \{\Phi(s) : s \in \mathcal{D}_i\}$. To compress $\mathcal{E}_i$ into a compact and representative set, we apply k-means clustering to obtain k cluster centers $\mathbf{p}_i = \{\mathbf{c}_1, \ldots, \mathbf{c}_k\}$. These centers serve as the prototype set for world model $\mathcal{W}_i$, capturing the core distributional structure of the domain.

Retrieval by Wasserstein distance. At test time, we similarly construct a prototype set $\mathbf{p}_{\text{tgt}}$ for the target environment from available observations. We then rank all world models by the Wasserstein distance between their prototype sets and the target prototype:

$$\mathbf{M}_{\text{ret}} = \underset{\mathcal{W}_i}{\text{top-k}}\; W(\mathbf{p}_i, \mathbf{p}_{\text{tgt}}), \tag{1}$$

where $W(\cdot,\cdot)$ denotes the n-dimensional Wasserstein distance. The Wasserstein distance is a natural measure of the discrepancy between prototype sets, as it respects the geometry of the embedding space and is robust to minor distributional shifts within a domain. The top-k world models with smallest Wasserstein distance are selected for implanting.

3.4. World-Wise Compound Attention

Once the relevant world models $\mathbf{M}_{\text{ret}} = \{\mathcal{W}_{(1)}, \ldots, \mathcal{W}_{(k)}\}$ are retrieved, we fuse their hidden representations with the frozen LLM using *world-wise compound attention*. The structure is illustrated in Figure 2.

Hook-based integration. WorMI registers forward hooks at selected layers of the LLM and each world model. At

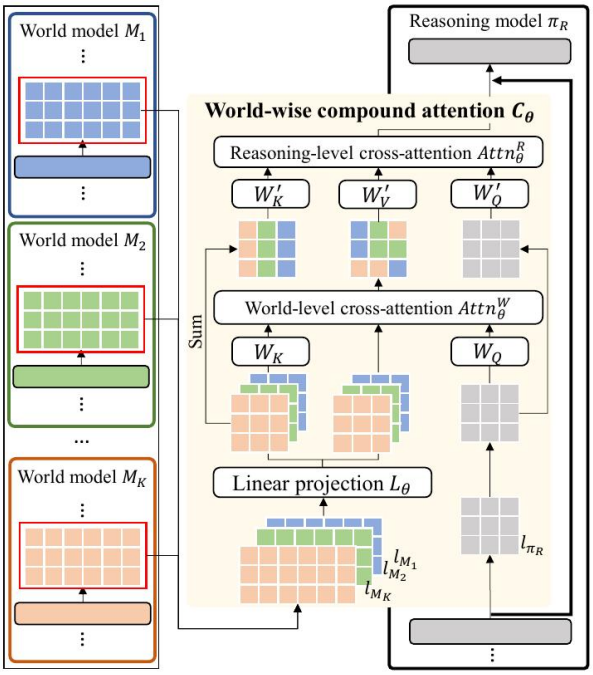


Figure 2: Overall structure of the world-wise compound attention. The mechanism operates in two stages: world-level cross-attention integrates multiple world model representations, and reasoning-level cross-attention aligns the fused world knowledge with the LLM's hidden states.

the hooked LLM layers, the hook receives the hidden state $\mathbf{h}_{\text{base}} \in \mathbb{R}^{B\times L\times d_{\text{base}}}$ as query, and collects the corresponding hidden states $\{\mathbf{h}^{(j)}_{\text{world}}\}_{j=1}^{k} \in \mathbb{R}^{k\times B\times L\times d_{\text{world}}}$ from the world models via extraction hooks.

Linear projection. A shared MLP projection $\mathbf{W}_{\text{proj}}$ maps world model hidden states from dimension d_{world} to d_{base}:

$$\tilde{\mathbf{h}}^{(j)}_{\text{world}} = \mathbf{W}_{\text{proj}}\, \mathbf{h}^{(j)}_{\text{world}}. \tag{2}$$

World-level cross-attention. We first integrate the k projected world representations via a *world-level cross-attention*. The query is the base hidden state $\mathbf{h}_{\text{base}}$ (reshaped per token), and the keys/values are the stack of world model representations:

$$\mathbf{K}_{\text{world}} = \sum_j \alpha_j \tilde{\mathbf{h}}^{(j)}_{\text{world}}, \tag{3}$$

$$\alpha_j = \text{softmax}\left(\frac{\mathbf{h}_{\text{base}}\,(\tilde{\mathbf{h}}^{(j)}_{\text{world}})^\top}{\sqrt{d_{\text{base}}}}\right), \tag{4}$$

producing an attention-weighted world representation $\mathbf{K}_{\text{world}}$.

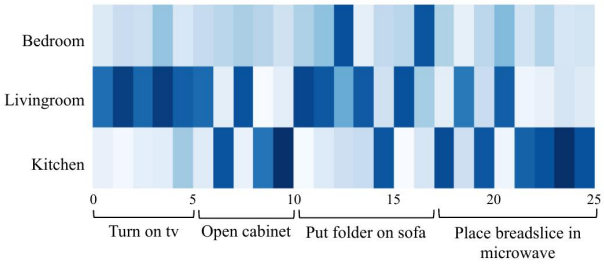


Figure 3: Visualization of the world-level attention map. The attention weights reveal which world models are most attended to at each token position, showing interpretable specialization by domain context.

Reasoning-level cross-attention. The fused world representation is then aligned with the LLM's hidden state via a *reasoning-level cross-attention*:

$$\mathbf{o}_{\text{attn}}, \omega = \text{CrossAttn}(\mathbf{h}_{\text{base}},\ \mathbf{K}_{\text{world}},\ \mathbf{K}_{\text{world}}), \tag{5}$$

$$\mathbf{h}_{\text{out}} = \mathbf{h}_{\text{base}} + \mathbf{W}_{\text{out}}\,\text{LayerNorm}(\mathbf{o}_{\text{attn}}). \tag{6}$$

The output $\mathbf{h}_{\text{out}}$ replaces the hooked layer's output, injecting world knowledge into the LLM's forward pass. Optionally, a self-attention layer further refines $\mathbf{h}_{\text{out}}$.

The world-level attention weights ω are interpretable and provide insight into which world models are most relevant for each token and position (Figure 3).

3.5. Meta-Learning for Rapid Adaptation

The compound attention composer (projection and attention parameters) is meta-learned using REPTILE (Nichol, 2018) across different subsets of world models drawn from the library. In each meta-iteration, we sample a random subset $\mathcal{S} \subset \{1, \ldots, N\}$ of world models, compute a few gradient steps on the composer using data from the selected domains, and update the meta-initialization towards the adapted parameters:

$$\phi \leftarrow \phi + \beta\,(\phi_{\mathcal{S}} - \phi), \tag{7}$$

where ϕ denotes the meta-initialized composer parameters, $\phi_{\mathcal{S}}$ is the adapted composer after inner-loop optimization on subset $\mathcal{S}$, and β is the meta learning rate. This procedure trains the composer to be rapidly specializable to any combination of world models, supporting the few-shot setting.

At test time, the retrieved world models are implanted by registering the extraction hooks on the world model layers and the compound attention hooks on the LLM layers. Removal is equally straightforward by deregistering the hooks. This enables *continual model implanting*: as the agent visits new domains, new world models are implanted without any LLM retraining, and obsolete ones are removed.

4. Experiments

4.1. Experimental Setup

Environments. We evaluate on two standard benchmarks for embodied instruction following:

- **VirtualHome** (Puig et al., 2018): a household simulator where agents execute daily activities described by natural language programs. We evaluate on seen/unseen task and scene combinations.
- **ALFWorld** (Shridhar et al., 2021): a text-based environment aligned with the ALFRED benchmark, requiring agents to complete multi-step household goals expressed in natural language.

Metrics. We report **Success Rate** (SR, ↑), the fraction of tasks completed successfully, and **Path Score** (PS, ↓), a measure of path length efficiency (lower is better). We use the 95% confidence interval over 5 random seeds.

Baselines. We compare against:

- **ZSP** (Huang et al., 2022a): zero-shot LLM planner with no fine-tuning.
- **LLM-FT**: LLM fine-tuned directly on task data.
- **LLM-Planner** (Song et al., 2023): few-shot grounded LLM planner.
- **SayCanPay** (Hazra et al., 2024): LLM planner with learned affordance (Can) and cost (Pay) functions.

Implementation. Unless otherwise stated, WorMI uses Llama-3.2-3B (AI@Meta, 2024) as the frozen reasoning LLM. World models are independently fine-tuned Llama-3.2-3B models on domain trajectory data. The compound attention uses 3 retrieved world models by default (see ablation in Section 4.4). The meta-learning follows the REPTILE algorithm (Nichol, 2018) with inner-loop optimization over domain subsets.

4.2. Zero-Shot Performance

Table 1 reports zero-shot performance across all environment configurations. WorMI consistently and substantially outperforms all baselines.

On VirtualHome in the most challenging setting (unseen tasks & unseen scenes), WorMI achieves 66.12% SR versus 45.71% for SayCanPay (the best baseline), a gain of +20.41%, while also achieving a substantially lower path score (15.17 vs. 19.04). On seen tasks and seen scenes, the advantage is even larger: 85.78% vs. 69.88%. On ALFWorld (unseen tasks & unseen scenes), WorMI achieves 51.67% SR vs. 39.66% (+12.01%) with a significantly lower path score (20.18 vs. 23.86). These results demonstrate that WorMI is highly effective at zero-shot generalization to novel environments and tasks by implanting relevant world knowledge at test time.

4.3. Few-Shot Performance

Table 2 reports few-shot performance across 1-shot, 5-shot, and 10-shot settings on the unseen tasks & unseen scenes configuration.

WorMI achieves substantially higher success rates than all baselines at every shot level. On VirtualHome, the average SR across shot levels is 77.51% for WorMI vs. 50.93% for SayCanPay (Δ+26.58%), with a lower average path score (11.90 vs. 16.88). On ALFWorld, WorMI averages 58.22% vs. 39.06% for SayCanPay (Δ+19.16%), with path score 15.46 vs. 22.82. Notably, WorMI in the 1-shot setting surpasses SayCanPay at 10-shot, demonstrating superior data efficiency due to the meta-learned composer.

4.4. Ablation Studies

Prototype-based retrieval and compound attention. Table 3 presents ablations on the two core components of WorMI using the zero-shot, unseen tasks & unseen scenes VirtualHome setting.

For the retrieval component, we compare:

- WorMI-**E**: no retrieval; all world models in the library are used without selection.
- WorMI-**R**: random retrieval without prototype matching.
- WorMI: full prototype-based retrieval with Wasserstein distance.

Prototype-based retrieval provides a 3.35% gain over random retrieval (66.12% vs. 62.04%) and a 17.49% gain over using all models (66.12% vs. 48.63%). This validates the importance of domain-aligned world model selection.

For the attention fusion mechanism, we compare:

- WorMI-**CONCAT**: concatenation of world model hidden states along the sequence dimension.
- WorMI-**ADD**: elementwise addition of world model hidden states.
- WorMI: world-wise compound attention.

Compound attention outperforms both simpler aggregation strategies by a wide margin (66.12% vs. 56.47% for ADD

Table 1: Zero-shot performance in VirtualHome and ALFWorld. We use the 95% confidence interval, using 5 random seeds.

Model	Seen Tasks & Seen Scenes		Seen Tasks & Unseen Scenes		Unseen Tasks & Unseen Scenes	
	SR (↑)	PS (↓)	SR (↑)	PS (↓)	SR (↑)	PS (↓)
Evaluation in VirtualHome						
ZSP	11.15%±0.65%	29.02±0.08	8.95%±1.13%	29.25±0.11	8.19%±0.33%	29.36±0.56
LLM-FT	58.55%±2.18%	17.37±0.34	53.42%±0.79%	17.70±0.23	42.82%±0.76%	20.79±0.17
LLM-Planner	35.67%±1.25%	27.15±0.13	28.55%±0.42%	27.04±0.12	21.45%±0.42%	27.73±0.05
SayCanPay	69.88%±2.32%	14.53±0.55	64.74%±0.87%	15.64±0.18	45.71%±0.59%	19.04±0.63
WorMI	**85.78%±0.45%**	**10.76±0.19**	**80.26%±1.02%**	**12.42±0.09**	**66.12%±0.80%**	**15.17±0.08**
Evaluation in ALFWorld						
ZSP	2.30%±0.21%	49.34±0.66	2.26%±0.09%	49.04±0.95	2.13%±0.09%	49.68±0.23
LLM+FT	45.72%±0.39%	14.63±1.35	29.26%±0.72%	38.49±2.87	29.40%±1.65%	46.84±0.15
LLM-Planner	17.73%±0.61%	32.09±2.90	18.63%±0.82%	40.50±2.57	12.31%±0.80%	46.33±1.68
SayCanPay	40.67%±1.24%	18.37±1.93	34.10%±1.17%	20.82±1.21	39.66%±1.43%	23.86±1.90
WorMI	**62.51%±1.65%**	**9.96±1.29**	**52.67%±1.39%**	**17.74±0.83**	**51.67%±2.23%**	**20.18±0.63**

Table 2: Few-shot performance in VirtualHome and ALFWorld. We use the 95% confidence interval, using 5 random seeds.

Model	1-Shot		5-Shot		10-Shot		Average	
	SR (↑)	PS (↓)	SR (↑)	PS (↓)	SR (↑)	PS (↓)	SR (↑)	PS (↓)
Evaluation in VirtualHome								
LLM-FT	42.35%±0.85%	20.46±0.15	47.22%±0.46%	16.82±0.29	51.45%±1.06%	16.82±0.39	47.01%±0.79%	18.03±0.28
LLM-Planner	24.63%±0.34%	26.19±0.31	29.80%±0.55%	26.98±0.08	33.49%±0.44%	26.60±0.16	29.31%±0.44%	26.59±0.18
SayCanPay	46.75%±1.02%	19.12±0.16	49.80%±1.11%	15.52±0.11	56.24%±0.79%	16.00±0.13	50.93%±0.97%	16.88±0.13
WorMI	**74.90%±1.57%**	**12.18±0.31**	**78.04%±0.34%**	**11.85±0.08**	**79.61%±0.99%**	**11.68±0.13**	**77.51%±0.98%**	**11.90±0.17**
Evaluation in ALFWorld								
LLM-FT	26.78%±1.20%	46.51±0.27	29.79%±1.36%	46.21±0.37	33.57%±0.98%	43.70±0.83	30.05%±1.18%	45.47±0.49
LLM-Planner	18.28%±1.06%	43.94±2.87	16.80%±1.46%	40.50±3.14	17.76%±1.25%	42.45±2.64	17.61%±1.26%	42.30±2.88
SayCanPay	40.94%±1.34%	21.58±1.14	36.70%±1.31%	22.15±3.55	39.54%±1.48%	24.74±3.48	39.06%±1.38%	22.82±2.72
WorMI	**51.50%±2.21%**	**21.46±3.96**	**58.69%±1.94%**	**13.14±1.00**	**64.46%±0.88%**	**11.79±0.81**	**58.22%±1.68%**	**15.46±1.92**

Table 3: Ablation study on prototype-based world model retrieval and world-wise compound attention

(a) Prototype-based world model retrieval

Model	SR (↑)	PS (↓)
WorMI-E	48.63%±0.25%	18.93±0.49
WorMI-R	62.04%±0.44%	16.96±0.20
WorMI	**66.12%±0.80%**	**15.17±0.08**

(b) World-wise compound attention

Model	SR (↑)	PS (↓)
WorMI-CONCAT	47.79%±2.39%	20.03±0.07
WorMI-ADD	56.47%±0.61%	17.43±0.30
WorMI	**66.12%±0.80%**	**15.17±0.08**

and 47.79% for CONCAT), confirming that the hierarchical attention is critical for effectively integrating multiple world models.

Number of world models. Table 4 analyzes the effect of the number of retrieved world models (WMs) on performance.

Performance peaks at 3 world models and degrades for larger numbers. With a single world model, the agent lacks complementary knowledge; too many models introduce irrelevant noise that the compound attention cannot fully suppress. The prototype-based retrieval naturally selects the optimal number per instance (defaulting to $k = 3$ in our experiments).

Table 4: Impact of world model scale (WMs)

Num. of WMs	SR (↑)	PS (↓)
1	42.75%±0.50%	20.54±0.02
2	62.94%±0.12%	17.21±0.20
3	66.12%±0.59%	15.17±0.08
4	61.96%±0.25%	16.68±0.34
6	48.23%±0.25%	18.93±0.49

LLM backbone. Table 5 examines robustness to the LLM backbone size.

WorMI consistently outperforms SayCanPay and LLM-Planner for all LLM sizes tested (Llama-3.2-1B, 3B, and 11B). Gains are largest for the 3B model (66.12% vs. 45.71% for SayCanPay) but remain substantial for the 1B model (49.65% vs. 42.22%). This shows that WorMI's gains are not specific to a particular LLM capacity, and

Table 5: Impact of LLMs on WorMI and baselines

LLM	Model	SR (↑)	PS (↓)
Llama-3.2-11B	LLM-Planner	51.11%±0.27%	18.33±0.08
	SayCanPay	49.28%±0.20%	18.88±0.10
	WorMI	**71.18%±0.62%**	**13.95±0.11**
Llama-3.2-3B	LLM-Planner	21.45%±0.42%	27.73±0.05
	SayCanPay	45.71%±0.59%	19.04±0.05
	WorMI	**66.12%±0.80%**	**15.17±0.08**
Llama-3.2-1B	LLM-Planner	9.93%±0.27%	27.37±0.11
	SayCanPay	42.22%±1.32%	21.34±0.33
	WorMI	**49.65%±0.83%**	**18.99±0.30**

Table 6: Performance for complex instruction scenarios

Model	SR (↑)	GC (↑)	PS (↓)
Long horizon instructions			
LLM-Planner	0.65%±0.51%	17.16%±0.58%	86.12±1.22
SayCanPay	6.54%±0.51%	39.22%±0.66%	57.87±0.28
WorMI	**19.61%±0.88%**	**49.35%±0.55%**	**53.15±0.70**
Multiple instructions			
LLM-Planner	1.31%±0.51%	29.02%±0.08%	85.17±0.76
SayCanPay	6.54%±0.52%	42.32%±0.41%	57.94±0.92
WorMI	**20.26%±1.34%**	**60.62%±0.89%**	**43.49±1.23**

that world model implanting can compensate for a weaker reasoning backbone.

4.5. Complex Instruction Scenarios

Table 6 evaluates WorMI on two challenging instruction types in VirtualHome, beyond standard single-step instructions.

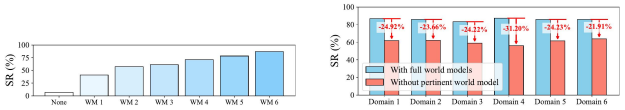
Long-horizon instructions. For tasks requiring many sequential steps, WorMI achieves 19.61% SR vs. 6.54% for SayCanPay (3× improvement), and goal completion (GC) of 49.35% vs. 39.22%. The world models provide crucial knowledge about multi-step affordances and object interactions that the frozen LLM lacks.

Multiple instructions. For tasks involving multiple concurrent instructions, WorMI achieves 20.26% SR vs. 6.54% for SayCanPay, and GC of 60.62% vs. 42.32%. Implanting multiple world models enables the agent to simultaneously reason over diverse sub-tasks, a setting where a single frozen LLM consistently fails.

4.6. Continual Model Implanting

We examine the behavior of WorMI as world models are sequentially implanted and removed, simulating a continual learning scenario.

Figure 4a shows that implanting world models sequentially generally improves task performance, as each new model contributes complementary domain knowledge. Figure 4b demonstrates that removing a pertinent world model degrades performance, while removing an irrelevant one has little or positive effect, validating the selectivity of the prototype-based retrieval mechanism. Together, these results confirm that WorMI supports continual, scalable knowledge composition without forgetting or interference between world models.



(a) Implanting WM1–WM6 sequentially.

(b) Effect of removing world models.

Figure 4: Performance for continual model implanting. (a) As world models WM1–WM6 are sequentially implanted, performance generally improves as complementary knowledge is accumulated. (b) Removing an irrelevant world model improves performance, while removing a pertinent one degrades it, confirming the selectivity of retrieval.

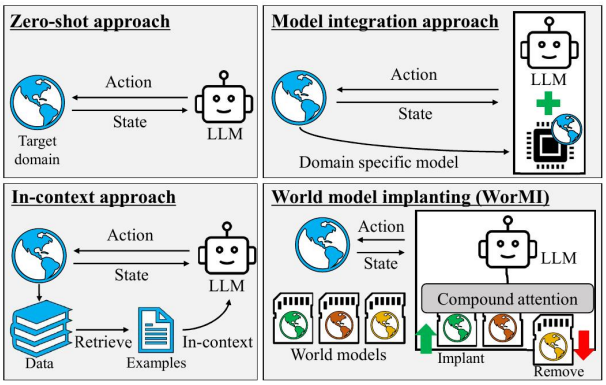


Figure 5: Concept of WorMI. At test time, domain-relevant world models are retrieved from a library and implanted into the frozen LLM policy via compound attention, enabling plug-and-play adaptation without retraining.

4.7. Concept Illustration

Figure 5 illustrates the core concept of WorMI: a frozen reasoning LLM is augmented with domain-specific world models retrieved from a library. Each world model captures a distinct aspect of environment knowledge, and the compound attention module fuses this knowledge seamlessly into the LLM's inference, providing adapted, grounded plans without any modification to the LLM parameters.

5. Conclusion

We presented WorMI, a test-time world model implanting framework for domain adaptation of LLM-based embodied agents. WorMI composes independently trained domain world models with a frozen LLM policy via prototype-based retrieval and world-wise compound attention. A meta-learned composer enables rapid specialization in few-shot settings. WorMI achieves substantial improvements over state-of-the-art baselines on VirtualHome and ALFWorld in zero-shot and few-shot settings, and demonstrates strong performance on long-horizon, multi-instruction, and continual implanting scenarios. By decoupling world knowledge from the reasoning model, WorMI provides a principled, scalable path to lifelong adaptation of embodied agents in open-ended environments.

Limitations and future work. The performance of WorMI depends on the quality and diversity of the world model library. If no similar world model exists in the library, retrieval may return poor matches. Future work could explore automatic world model creation from online observations, and extending WorMI to visual embodied environments where world models capture visual affordances.

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